ASSIGNMENT 1

1. What are the key features of Python?

Ans

```
1) dynamic language because we dont have to mention variable type
while declaring and defining
2) it is interpreter
3) high level language
4) extensive library
5) it is object oriented language
6) easy to learn and read
```

2. What are the Data Types in Python?

Ans -

```
int, float, str, bool, list, tuple, dictionary, set, frozenset,
range
```

3. What are local variables and global variables in Python?

Ans-

```
Local variable: Defined inside a function and accessible within
that function.
Global variable: Defined outside functions and accessible
throughout the module, subject to scope rules.
```

4. How do you write comments in Python? And Why Comments are important?

Ans-

```
by using # in front of any line, interpreter consider that line as
comment. we use comment because it make program readable to other
developers also.
```

5. How to comment on multiple lines in Python?

Ans-

by using triple quotes like ''' or """, we can comment multiple line by using multiline string or # infrony of all line

6. What do you mean by Python literals?

Ans-

python literals are those which is raw data present in source code. they fixed value that we don't need to compute to get. ex => 3.14

7. What are the different ways to assign value to variables?

Ans

1 Assignment operator is used to assign values in variable example name = "Deepti"
2 chained - a = b = c = 10
3 Multiple assignment - a,b = 10,11
4 unpacked assignment - a,b = list
5 Augmented assignment - a += 10

8. What are the Escape Characters in Python?

Ans-

escape character is used to perform specific task or if you want to print special character in string then also we use it

9. What are the different ways to perform string formatting? Explain with an example.

Ans

1) f string - print(f'hello {name}') name is variable which we used to insert name variable in string
2) format() - "hello {}".format(name) - here {} is placeholder

```
where name variable can be substituted
3) % formatting - "hello %s" % (name) - here %s will be replaced by
name
4) template     -  t = Template("hello {name}")
                   t.substitute(name=name)
```

10. Write a program to print every character of a string entered by the user in a new line using a loop

In [3]:
```
s = input("Enter String")
s = ' '.join(s)
for i in s.split(" "):
    print(i)
```

```
h
e
l
l
o
```

11. Write a program to find the length of the string "machine learning" with and without using the len function.

In [9]:
```
st = "machine learning"
print("output with len function - " + str(len(st)))
count = 0
st =  "-".join("machine learning")

for i in st.split("-"):
    count += 1
print("output without len function - "+ str(count))
```

```
output with len function - 16
output without len function - 16
```

12. Write a program to check if the word 'orange' is present in the "This is orange juice".

In [11]:
```
st = "This is orange juice"
word = 'orange'

if word in st:
    print("yes orange is present in string ")
else:
    print("No orange is not present in string ")
```

```
yes orange is present in string
```

13. Write a Python program to count the number of vowels, consonants, digits and special characters in a string

In [20]:
```
st = input("Enter the string")
# vowels
st.lower()
v_count = 0
for i in "aeiou":
    if i in st:
        v_count += st.count(i)
# consonant
c_count = 0
i = 'a'
while i>='a' and i<='z':
    if i in st:
        c_count += st.count(i)
    i = chr(ord(i) + 1)
# digit
d_count = 0
i = 0
while i>=0 and i<=9:
    if str(i) in st:
        d_count += st.count(str(i))
    i +=1

#special
s_count = len(st) - v_count - c_count - d_count
print(f'vowel count                 - {v_count}')
print(f'consonant count             - {c_count}')
print(f'digit count                 - {d_count}')
print(f'special character count     - {s_count}')
```

```
vowel count                 - 6
consonant count             - 9
digit count                 - 6
special character count     - 5
```

In [ ]:
```
```

14. Write a Python program to count Uppercase, Lowercase, special characters, and numeric values in a given string.

In [21]:
```
st = input("Enter the string")
up = 0
lw = 0
sc = 0
num = 0
for i in st:
```

```
    if i.isupper():
        up += 1
    elif i.islower():
        lw += 1
    elif i.isnumeric():
        num += 1
    else:
        sc += 1

print(f'upper case count        - {up}')
print(f'lower case count        - {lw}')
print(f'numeric count           - {num}')
print(f'special character count - {sc}')
```

```
upper case count        - 2
lower case count        - 16
numeric count           - 8
special character count - 10
```

In [ ]:

15. Write a program to make a new string with all the consonants deleted from the string "Hello, have a good day".

In [4]:
```
st = "Hello, have a good day"
res = ""
for i in st:
    if i not in 'aeiou':
        if i.lower() < 'a' or i.lower() > 'z':
            res += i
    else:
        res += i
print(res)
```

```
eo, ae a oo a
```

In [ ]:

16. Write a Python program to remove the nth index character from a non-empty string.

In [6]:
```
st = input("Enter a string ")
index = int(input("Enter the index whose value to be deletes"))
res = ""
count = 0
for i in st:
    if count != index:
        res += i
```

```
    count += 1
print("output -  " + res )
```

```
output -  helo
```

In [ ]:

17. Write a Python program to change a given string to a new string where the first and last characters have been exchanged.

In [7]:
```
st = input("Enter a string ")
print(st[-1] + st[1:-1] + st[0])
```

```
iello Deepth
```

In [ ]:

18. Write a Python program to count the occurrences of each word in a given sentence.

In [9]:
```
st = input('Enter the string')
l = st.split()
s = set(l)
d = {}
for i in s:
    d[i] = l.count(i)

print(d)
```

```
{'!': 1, 'hello': 2, 'jane': 1, 'girls': 1}
```

In [ ]:

19. How do you count the occurrence of a given character in a string?

In [10]:
```
st = input('Enter the string')
l = list(st)
s = set(l)
d = {}
for i in s:
    d[i] = l.count(i)

print(d)
```

```
{'h': 1, 'o': 1, 'e': 1, 'l': 2}
```

In [ ]:

20. Write a program to find the last 10 characters of a string.

In [12]:
```
st = input('Enter the string')
if len(st) >= 10:
    print(st[-10:])
else:
    print(st)
```

```
Assignment
```

In [ ]:
```

```

21. WAP to convert a given string to all uppercase if it contains at least 2 uppercase characters in the first 4 characters.

In [5]:
```
st = input("Enter the String")
count = 0
for i in st[0:4]:
    if i.isupper():
        count += 1
if count >= 2:
    print("output - " + st.upper())
else:
    print("output - " + st)
```

```
output - HELLO
```

In [ ]:
```

```

22. Write a Python program to remove a new line from a string.

In [8]:
```
st = '''hello deepti.
she is an enginner.
she is doing great.
'''
st.replace("\n",'')
```

Out[8]:
```
'hello deepti.she is an enginner.she is doing great.'
```

In [ ]:
```

```

23. Write a Python program to swap commas and dots in a string

- Sample string: "32.054,23" >> Expected Output: "32,054.23"

In [11]:
```
st = input("Enter the string ")
res = ""
for i in st:
    if ',' == i:
        res += "."
    elif '.' == i:
        res += ","
    else:
        res += i

print("output - " + res)
```

```
output - hello, deepti . hello. class,
```

In [ ]:
```

```

24. Write a Python program to find the first repeated character in a given string

In [16]:
```
st = input("Enter the string ")
counts = 0
for i in st:
    if st.count(i) > 1:
        print("output - " + i)
        break
    counts += 1
if counts == len(st):
        print("output - " + str(-1))
```

```
output - l
```

In [ ]:
```

```

25. Write a Python program to find the second most repeated word in a given string

In [6]:
```
st = input("Enter the string ")
counts = 0
first = ""
second = ""
c = 0
for i in set(st.split()):
    if counts < st.count(i):
        c = counts
        counts = st.count(i)
        second = first
        first = i
    elif c < st.count(i) and  st.count(i) < counts:
        second = i
```

```
        c = st.count(i)
print("output - "+ second )
```

```
1
2
output - odd
```

In [ ]:

26. Python program to Count Even and Odd numbers in a string.

In [12]:
```
st = input("Enter the string ")
e_count = 0
o_count = 0
for i in st.split():
    if i.isdigit() and int(i)%2 == 0:
        e_count += 1
    elif i.isdigit() and int(i)%2 != 0:
        o_count += 1
print(f"even count = {e_count}\nodd count = {o_count}")
```

```
even count = 1
odd count = 1
```

In [ ]:

27. How do you check if a string contains only digits?

In [18]:
```
st = input("Enter string - ")
if st.isdigit():
    print("yes")
else:
    print("No")
```

```
yes
```

In [ ]:

28. How do you remove a given character/word from String?

In [20]:
```
st = input("Enter string - ")
word = input("Enter word to be removed")

st.replace(word,"")
```

Out[20]: 'jfsoidfjoi'

In [ ]:

29. Write a Python program to remove the characters which have odd index values of a given string

In [23]:
```
st = input("Enter string - ")
count = 0
res = ""
for i in st:
    if count%2 == 0:
        res += i
    count += 1
print("output - " + res)
```

```
output - hpy
```

In [ ]:

30. Write a Python function to reverse a string if its length is a multiple of 5

In [28]:
```
st = input("Enter string - ")
if len(st) % 5 == 0:
    print("output - "+ st[::-1])
```

```
output - tppah
```

In [ ]:

31. Write a Python program to format a number with a percentage(0.05 >> 5%)

In [32]:
```
st = float(input("Enter number - "))
print("output - "+ str(round(st * 100)) + "%")
```

```
output - 5%
```

In [ ]:

32. Write a Python program to reverse words in a string

In [33]:
```
st = input("Enter String - ")
res = ""
for i in st.split():
    res += i[::-1]
    res += " "
```

```
res.rstrip()
print("output - "+res)
```

```
output - yppah yad
```

In [ ]:

33. Write a Python program to swap cases of a given string

In [35]:
```
st = input("Enter String - ")
print("output - "+st.swapcase())
```

```
output - HAPPy
```

In [ ]:

34. Write a Python program to remove spaces from a given string

In [39]:
```
st = input("Enter String - ")
res = ""
for i in st.split():
    res += i
res.strip()
print("output - "+ res)
```

```
output - happyday
```

In [ ]:

35. Write a Python program to remove duplicate characters of a given string

In [43]:
```
st = input("Enter String - ")
l = set(st)
d = {}
res = ""
for i in l:
    d[i] = st.count(i)
for i in st:
    if i in d:
        res+= i
        del d[i]

print("output - "+ res)
```

```
output - hapy bir
```

In [ ]:

36. Write a Python Program to find the area of a circle

In [46]:
```
st = int(input("Enter radius - "))
pi = 3.14
print("output - "+ str(pi * st *st))
```

```
output - 78.5
```

In [ ]:

37. Write a Python Program to find the Sum of squares of first n natural numbers

In [48]:
```
st = int(input("Enter n value - "))
sums = 0
for i in range(1,st+1):
    sums += i * i
print("output - " + str(sums))
```

```
output - 55
```

In [ ]:

38. Write a Python Program to find the Sum of cubes of first n natural numbers

In [49]:
```
st = int(input("Enter n value - "))
sums = 0
for i in range(1,st+1):
    sums += i * i *i
print("output - " + str(sums))
```

```
output - 225
```

In [ ]:

39. Write a Python Program to find simple interest and compound interest

In [52]:
```
p = float(input("Enter principal value - "))
r = float(input("Enter rate value - "))
t = float(input("Enter time value - "))
print("output SI - " + str((p*r*t)/100))
print("output CI - " + str(p *((1 + r/100)**t)))
```

```
output SI - 5000.0
output CI - 16288.94626777442
```

In [ ]:

40. Write a Python program to check whether a number is Prime or not

In [66]:
```
p = int(input("Enter a number - "))
j = 0
for i in range(2,int(p**0.5)+1):
    if p%i == 0:
        j = 1
        break
if j != 1:
    print("output - prime ")
else:
     print("output - not prime ")
```

```
output - prime
```

In [ ]:

In [ ]: